Predictive Credit Scoring Model

Abstract

This report presents the design, implementation, and empirical validation of an end-to-end machine learning system developed to predict individual default risk using past consumer financial profiles. Using Kaggle's historical data of 150,000 credit applicants, we systematically addressed severe class imbalance—characterized by a 93.3% non-default to 6.7% default ratio—and executed robust preprocessing pipelines to neutralize operational anomalies. An ensemble Random Forest Classifier was trained under a cost-sensitive learning framework to prioritize default sensitivity. The final model demonstrated strong discriminative capacity, yielding an Area Under the Receiver Operating Characteristic Curve (ROC-AUC) of 0.8608. While overall accuracy registered at 77%, the system achieved a high-risk default recall of 80%, successfully catching the vast majority of potential defaults. This report evaluates the underlying methodology and discusses the mathematical and strategic trade-offs of the model's classification metrics.

1. Introduction and Project Motivation

Credit scoring systems serve as a core risk mitigation tool in modern retail banking. The primary challenge in underwriting is to accurately differentiate between creditworthy applicants and potential defaulters. Imbalances in this assessment yield distinct operational costs:

- **Type I Error (False Positive):** Denying credit to a viable applicant. This leads to customer friction and immediate loss of interest-earning assets.
- **Type II Error (False Negative):** Approving credit for an applicant who eventually defaults. This causes a direct write-off of the outstanding principal, severely eroding institutional capital.

In consumer lending, Type II errors are orders of magnitude more financially damaging than Type I errors. Therefore, standard predictive models optimized purely for accuracy are insufficient. This project constructs an optimized, cost-sensitive classification framework to minimize credit default rates while maintaining sustainable credit approval flows.

2. Preprocessing and Data Quality Assurance

Raw financial transaction records often contain extreme statistical anomalies and missing records due to system errors or structural limitations. To prepare the training matrix, we executed a robust, future-proof preprocessing pipeline:

- **Imputation of Missing Fields:** * Missing elements in `MonthlyIncome` were imputed using the median (X_{median}) rather than the mean to prevent skewness from multi-million-dollar high-income outliers.
  - Missing entries in `NumberOfDependents` were resolved utilizing the column's statistical mode (0).
- **Outlier Capping Logic:**
  - The variable `RevolvingUtilizationOfUnsecuredLines`, representing the ratio of outstanding debt to credit limits, contained extreme errors exceeding $50,000$. Since a standard utilization ratio should scale between 0 and 1.0, a threshold of 2.0 was established; any values beyond this limit were capped at 1.0 (representing full utilization).
  - Extreme delinquency counts (where indicators reached placeholders of 96 or 98) were capped using localized column medians to prevent decision trees from forming highly distorted splitting thresholds.

3. Machine Learning Architecture and Strategic Split

To prevent data leakage and establish a reliable testing framework, the processed data was partitioned into an 80% training set and a 20% testing set.

Because defaults constitute a scarce minority (6.7%), we applied a **stratified split** matching the exact target distribution in both subsets. This preserves the structural complexity of the risk pool across training and testing phases.

Algorithm Selection

A **Random Forest Classifier** was selected for its resilience to collinear features and its ability to map non-linear relationships. To counter the severe class imbalance, the classifier was configured with:

- `class_weight='balanced'` : This mathematically adjusts the cost function during training, assigning a higher loss penalty to misclassifications of the minority (defaulting) class.
- `max_depth=6` : Restricting the tree depth prevents individual decision boundary lines from overfitting on minority noise, ensuring strong generalization on unseen test profiles.

4. Empirical Performance Evaluation

The model's classification performance on the unseen test set ($30,000$ consumer records) yielded the following results:

Classification Report Analysis

Class Indicator	**Precision**	**Recall**	**F1-Score**	**Support**
Good Credit (0)	0.98	0.77	0.86	27,995
High Risk (1)	0.20	0.80	0.32	2,005
Overall Accuracy			0.77	30,000
Macro Average	0.59	0.79	0.59	30,000
Weighted Average	0.93	0.77	0.83	30,000

Deep Dive into the Metrics

1. **The Accuracy Paradox (77%):** While an overall accuracy of 77% is lower than the baseline baseline rate of predicting "no default" (93.3%), it represents a much safer model. A naive model predicting 100% approval achieves 93.3% accuracy but fails to catch any defaults, resulting in massive credit losses.
2. **High-Risk Default Recall (80%):** The model achieved a Recall of 0.80 for Class 1. This means the model successfully identified and flagged 80% **of all actual defaulting borrowers** in the test set. For risk mitigation, this is the most critical benchmark metric.
3. **High-Risk Precision (20%):** The precision for Class 1 is 0.20. This indicates that among all applications flagged as "High Risk", 20% ultimately defaulted, while the remaining 80% did not. In retail banking, this low precision is an expected, acceptable trade-off for protecting capital. Rejecting a portion of creditworthy applicants is much less costly than approving bad loans that default on their entire principal.
4. **ROC-AUC Score (0.8608):** An ROC-AUC score of 0.8608 indicates excellent discriminative power. It proves that there is an 86.08% probability that the model will rank a randomly chosen defaulting applicant as riskier than a randomly chosen creditworthy applicant.

5. Institutional Implications and Conclusion

The empirical findings validate the integration of this predictive model into modern lending environments. By shifting the decision threshold, institutions can tune this model to match their specific risk appetites:

- **Conservative Underwriting:** Prioritizes maximizing Recall ($80\%+$) to completely insulate capital reserves from default shocks, accepting lower precision.
- **Aggressive Underwriting:** Relaxes the threshold to improve precision, unlocking higher credit approval volumes and interest generation in low-risk macroeconomic climates.

In conclusion, moving away from standard accuracy optimization to a cost-weighted Random Forest classifier provides a balanced, mathematically sound underwriting engine. The model successfully isolates high-risk credit seekers, preserving financial stability while automating credit scoring at scale.